

of the CPU in mobile devices," adds Abhishek.

"An increase in clock speed or bus-width increases power consumption in direct proportion. Therefore design the system to run as slow as practical in all the operating modes that you have decided," advises T. Anand, MD, Knewron.

Next, check whether the processor is static or dynamic. "Dynamic core processors are not usually made for low-power applications as they don't respond well to slow or stopped clocks. If the datasheet says $f_{osc} = 0$, you can slow down as much as you want," Anand explains.

These idle modes ensure that the system merely sips power when not in use while at the same ensuring sufficient power for intensive number-crunching sessions. The silicon itself is built with a technology that ensures ultra-low power consumption.

"Engineers have now implemented an effective clock circuit strategy to ensure that unused processes can be shut down and consumption kept to a minimum. Moreover, in our pursuit of better performance, we have moved away from proprietary cores towards ARM cores. This has enabled us to have different products, each targeted towards high performance, standard low power and even ultra-low power," explains Affa Au, director-marketing, South Asia, Fujitsu Semiconductor Asia (Pte) Ltd.

"While slowing down strategy is going to save power, it sure will affect computing capability negatively. Having clock speed control on-the-go can help a lot in such cases, where device remains slow or static in power saving/down mode and speeds up only when necessary," adds Anand.

Energy efficiency requires more of an evolutionary than revolutionary approach. "In a portable environment such as a handheld industrial test gadget, it is highly desirable for the device to last a complete shift on a single charge and so enabling low-power 'drowse' modes when not in use is sensible. If it takes the device 0.5 second to come back up to operational speed, that is likely acceptable. Other industrial devices that have real-time control functions like a robot are more safety-oriented. The controller part probably consumes a small percentage of the energy use compared to the motor drives and so controller power saving is less desirable," explains Brian Brown, vice president of marketing, embedded computing, Emerson Network Power.

At times it also becomes necessary to remove certain popular features or at least replace them with good alternatives.

"As people tried to make devices cheaper and low-power especially on the consumer side, all that the traditional Adobe Flash provided was a heavier stack to implement in embedded processors. It runs about 2.5 million lines of code on the Flash player, and that does take quite a bit of computation power. This, along with the resulting impact on battery life, is also a concern. It was one of the main reasons for Adobe to discontinue support for Flash on the embedded side and support HTML 5 that is gaining a lot of acceptance today," explains Mrinmoy Purkayastha, associate vice president-marketing, and Somenath Nag, director-business development and marketing, ALTEC Calsoft Labs.

Since the Internet of Things is

already here, designers also need to consider the penalties of having an always-connected device.

Pratul Schroff, MD of eInfoChips, advises, "In these cases, one thing that is being done is to scale down the processing speed or the clock speed, so that they only need low power. Typically, the operating frequency in a modern mobile device is 1 GHz but we scale it down to 20 MHz at other instances where processing power is enough to handle the data but the power consumption remains controlled. Thus we run at 1/50th of its standard speed, and by doing so reap the benefits of lower power consumption."

Often, the connected peripherals are also to blame for inadequate battery life. Milind Gandhe, associative vice president, semiconductor business line, explains how Sasken Communication Technologies ensures low power consumption in its designs: "Power consumption of the device is directly proportional to the leakage current. So we make sure that a lot of our designs are basically centered at minimising leakage current. A key challenge to reduce power consumption in these systems is to ensure that the power drawn by the peripherals is low."

You can influence your system's uptime

It's time designers stopped pointing to battery technology as the reason for awfully low uptime on their devices. Like Stephen R. Covey once said, "Proactive people focus their efforts on their circle of influence." Designing an efficient circuit is under your influence. So go make it count. ●

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